Lars van der Laan

Postdoctoral Scholar, Stanford Management Science & Engineering
Stanford Causal Science Center Fellow
larsvanderlaan.github.io | vdlaan@stanford.edu | Google Scholar

Education

University of Washington 09/2021–06/2026
Ph.D. in Statistics, June 2026; Advisors: Marco Carone and Alex Luedtke
Dissertation: Improved Debiased Machine Learning Through Optimization and Calibration

University of California, Berkeley 08/2019–05/2020
M.A. in Statistics, May 2020

University of Groningen, Netherlands 09/2016–07/2019
B.Sc. in Mathematics and Physics, cum laude, July 2019

Appointments

Stanford University 07/2026–Present
Postdoctoral Scholar, Management Science & Engineering (V. Syrgkanis; Netflix Research)

Stanford Causal Science Center 07/2026–Present
Fellow

Netflix 09/2024–06/2026
Graduate Research Fellow (with A. Bibaut and N. Kallus)

Netflix 06/2024–08/2024
Machine Learning Research Intern

UC Berkeley, Computational Precision Health 09/2023–06/2024
Visiting Student Researcher (with A. Alaa)

University of Washington 09/2022–06/2023
Teaching Assistant, Ph.D. Theory Sequence

University of Washington 09/2021–06/2026
Research Assistant, Department of Statistics

Fred Hutchinson Cancer Center 09/2020–06/2024
Research Assistant and Research Intern (with P. Gilbert)

Genentech 06/2020–08/2020
Research Intern (with J. Levy)

UC Berkeley, Environmental Health Sciences 09/2020–08/2021
Research Assistant (with A. Cardenas)

Research Interests

Causal inference and missing data; semiparametric statistics and debiased machine learning; statistical learning theory and heterogeneous treatment effects; policy learning, reinforcement learning, and dynamic experiments; predictive inference, uncertainty quantification, and calibration.

Awards & Fellowships

- Stanford Causal Science Center Fellowship, 2026–Present
- Netflix Graduate Research Fellowship, 2024–2026 (Mentors: Aurélien Bibaut and Nathan Kallus)
- Best Paper Award, Conference on Causal Learning and Reasoning (CLeaR), 2025
- Tom Ten Have Poster Award, Honorable Mention, American Causal Inference Conference, 2025
- University of Washington Statistics Department Fellowship, 2021

Research Publications

Preprints and Working Papers

[1] L. I. B. Andrews, **L. van der Laan**, and P. B. Gilbert. "Targeted Maximum Likelihood Estimation of Vaccine Effectiveness and Immune Correlates in Test-Negative Design Studies with Missing Data". In: *arXiv preprint arXiv:2605.21793* (2026).

[2] **L. van der Laan**. "A Researcher's Guide to Empirical Risk Minimization". In: *arXiv preprint arXiv:2602.21501* (2026).

[3] **L. van der Laan** and N. Kallus. "Bellman Calibration for Marginalized Importance Weighting in Offline Reinforcement Learning". In: *arXiv preprint arXiv:2608.24858* (2026).

[4] **L. van der Laan** and N. Kallus. "Fitted Occupancy-Ratio Evaluation without Bellman Completeness". In: *arXiv preprint arXiv:2607.05375* (2026).

[5] **L. van der Laan**, V. Syrgkanis, and N. Kallus. "Model Selection for Off-Policy Evaluation via Pairwise Bellman Validation". In: *arXiv preprint arXiv:2610.35748* (2026).

[6] **L. van der Laan**, A. Bibaut, and N. Kallus. "Efficient Inference for Inverse Reinforcement Learning and Dynamic Discrete Choice Models". In: *arXiv preprint arXiv:2512.24407* (2025).

[7] **L. van der Laan**, A. Bibaut, N. Kallus, and A. Luedtke. *Automatic Debiased Machine Learning for Smooth Functionals of Nonparametric M-Estimands*. 2025.

[8] **L. van der Laan**, D. Hubbard, A. Tran, N. Kallus, and A. Bibaut. *Semiparametric Double Reinforcement Learning with Applications to Long-Term Causal Inference*. 2025.

[9] L. van der Laan and N. Kallus. "Bellman Calibration for V-Learning in Offline Reinforcement Learning". In: *arXiv preprint arXiv:2512.23694* (2025).

[10] **L. van der Laan** and N. Kallus. "Fitted Q-Evaluation without Bellman Completeness via Occupancy Weighting". In: *arXiv preprint arXiv:2512.23805* (2025).

[11] **L. van der Laan** and N. Kallus. "Soft Fitted Q-Iteration without Bellman Completeness: Occupancy Reweighting and Temperature Annealing". In: *arXiv preprint arXiv:2512.23927* (2025).

[12] **L. van der Laan**, N. Kallus, and A. Bibaut. *Nonparametric Instrumental Variable Inference with Many Weak Instruments*. 2025.

[13] **L. van der Laan**, M. Carone, and A. Luedtke. "Combining T-learning and DR-learning: a framework for oracle-efficient estimation of causal contrasts". In: *arXiv preprint arXiv:2402.01972* (2024).

[14] **L. van der Laan**, A. Luedtke, and M. Carone. "Doubly robust inference via calibration". In: *arXiv preprint arXiv:2411.02771* (2024).

[15] S. Dutta, H. Wei, **L. van der Laan**, and A. Alaa. "Estimating Uncertainty in Multimodal Foundation Models using Public Internet Data". In: *arXiv preprint arXiv:2310.09926* (2023).

[16] **L. van der Laan**, M. Carone, A. Luedtke, and M. van der Laan. "Adaptive debiased machine learning using data-driven model selection techniques". In: *arXiv preprint arXiv:2307.12544* (2023).

[17] **L. van der Laan** and P. B. Gilbert. "Semiparametric logistic regression for inference on relative vaccine efficacy in case-only studies with informative missingness". In: *arXiv preprint arXiv:2303.11462* (2023).

[18] M. van der Laan, Z. Wang, and **L. van der Laan**. "Higher order targeted maximum likelihood estimation". In: *arXiv preprint arXiv:2101.06290* (2021).

Conference Papers

[19] Z. Liang, **L. van der Laan**, and A. Alaa. "Hybrid Meta-learners for Estimating Heterogeneous Treatment Effects". In: *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics* (*AISTATS*). Vol. 300. Proceedings of Machine Learning Research. PMLR, 2026, pp. 3385–3393.

[20] **L. van der Laan**, N. Kallus, and A. Bibaut. "Inverse Reinforcement Learning with Just Classification and a Few Regressions". In: *Advances in Neural Information Processing Systems 39* (*NeurIPS 2026*). 2026.

[21] **L. van der Laan** and M. van der Laan. "Calibeating Prediction-Powered Inference". In: *Advances in Neural Information Processing Systems 39* (*NeurIPS 2026*). 2026.

[22] **L. van der Laan** and A. Alaa. "Generalized Venn and Venn-Abers Calibration with Applications in Conformal Prediction". In: *Proceedings of the 42nd International Conference on Machine Learning* (*ICML*). Vol. 267. Proceedings of Machine Learning Research. Vancouver, Canada: PMLR, 2025.

[23] **L. van der Laan**, Z. Lin, M. Carone, and A. Luedtke. "Stabilized Inverse Probability Weighting via Isotonic Calibration". In: *Proceedings of the Fourth Conference on Causal Learning and Reasoning* (*CLeaR*). Vol. 275. Proceedings of Machine Learning Research. PMLR, 2025, pp. 139–173.

[24] **L. van der Laan** and A. M. Alaa. "Self-Calibrating Conformal Prediction". In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems* (*NeurIPS*) (2024).

[25] **L. van der Laan**, E. Ulloa-Pérez, M. Carone, and A. Luedtke. "Causal isotonic calibration for heterogeneous treatment effects". In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. Vol. 202. Proceedings of Machine Learning Research. Honolulu, Hawaii, USA: PMLR, 2023, pp. 34831–34854.

Journal Articles

[26] H. Janes, Y. Fong, Y. Huang, D. Benkeser, E. J. Kelly, I. Hirsch, A. M. Stanley, T. Villafana, et al. "Correlates of Severe and Delta COVID-19 in a Phase 3 Trial of the AZD1222 Vaccine". In: *npj Vaccines* 11.1 (2026), p. 120.

[27] M. van der Laan, S. Qiu, J. M. Tarp, and L. van der Laan. "Adaptive-TMLE for the average treatment effect based on randomized controlled trial augmented with real-world data". In: *Journal of Causal Inference* 14.1 (2026), p. 20240025.

[28] R. Wyss, B. B. Hansen, G. Hahn, **L. van der Laan**, and J. K. Lin. "Tuning LASSO Models for Propensity Score Weighting and Using Synthetic Negative Control Exposures for Residual Bias Detection". In: *Statistics in Medicine* 45.8–9 (2026), e70503.

[29] L. I. B. Andrews, M. E. Halloran, K. M. Neuzil, **L. van der Laan**, Y. Huang, J. Andriesen, M. Patel, L. H. Fisher, H. Janes, N. Rouphael, et al. "Evaluating the Test-Negative Design for COVID-19 Vaccine Effectiveness Using Randomized Trial Data: A Secondary Cross-Protocol Analysis of 5 Randomized Clinical Trials". In: *JAMA Network Open* 8.5 (2025), e2512763.

[30] G. Baek, M. Kim, M. Lee, S. O'Connor, L. Held, **L. van der Laan**, and R. D. Cassaday. "Retrospective Review of the Toxicities and Change in Dosing Patterns for Pegaspargase in Patients with Acute Lymphoblastic Leukemia/Lymphoma and T-cell Lymphoma". In: *Journal of Oncology Pharmacy Practice* 31.4 (2025), pp. 534–544.

[31] A. Kenny, **L. van der Laan**, P. B. Gilbert, and M. Carone. "Inference on Controlled Effects for Assessing Immune Correlates of Protection Based on a Cox Model". In: *Statistics in Medicine* 44.28–30 (2025), e70347.

[32] D. Khodasevich, N. Holland, **L. van der Laan**, and A. Cardenas. "A SuperLearner-Based Pipeline for the Development of DNA Methylation-Derived Predictors of Phenotypic Traits". In: *PLOS Computational Biology* 21.2 (2025), e1012768.

[33] A. Luedtke, Y. Fong, **L. van der Laan**, F. Heng, Y. Huang, Y. Lu, C. Yu, L. N. Carpp, S. Roels, M. Le Gars, et al. "Immune Correlates Analysis of Antibody Responses against SARS-CoV-2 Variants in the ENSEMBLE Vaccine Efficacy Trial". In: *iScience* 28.11 (2025).

[34] Z. Wang, **L. van der Laan**, M. Petersen, T. Gerds, K. Kvist, and M. van der Laan. "Targeted maximum likelihood based estimation for longitudinal mediation analysis". In: *Journal of Causal Inference* 13.1 (2025), p. 20230013.

[35] A. Kenny, J. van Duijn, O. Dintwe, J. Heptinstall, R. Burnham, S. Sawant, L. Zhang, D. Mielke, et al. "Immune Correlates Analysis of the Imbokodo (HVTN 705/HPX2008) Efficacy Trial of a Mosaic HIV-1 Vaccine Regimen Evaluated in Southern African People Assigned Female Sex at Birth: A Two-Phase Case-Control Study". In: *EBioMedicine* 108 (2024), p. 105320.

[36] B. Zhang, Y. Fong, J. Fintzi, E. Chu, H. E. Janes, A. Kenny, M. Carone, D. Benkeser, **L. van der Laan**, W. Deng, et al. "Omicron COVID-19 Immune Correlates Analysis of a Third Dose of mRNA-1273 in the COVE Trial". In: *Nature Communications* 15.1 (2024), p. 7954.

[37] D. Benkeser, Y. Fong, H. E. Janes, E. J. Kelly, I. Hirsch, S. Sproule, A. M. Stanley, J. Maaske, et al. "Immune Correlates Analysis of a Phase 3 Trial of the AZD1222 (ChAdOx1 nCoV-19) Vaccine". In: *npj Vaccines* 8.1 (2023), p. 36.

[38] D. Benkeser, D. C. Montefiori, A. B. McDermott, Y. Fong, H. E. Janes, W. Deng, H. Zhou, C. R. Houchens, et al. "Comparing Antibody Assays as Correlates of Protection against COVID-19 in the COVE mRNA-1273 Vaccine Efficacy Trial". In: *Science Translational Medicine* 15.692 (2023), eade9078.

[39] P. Collender, A. K. Bozack, S. Veazie, J. C. Nwanaji-Enwerem, **L. van der Laan**, K. Kogut, C. Riddell, B. Eskenazi, N. Holland, J. Deardorff, et al. "Maternal Adverse Childhood Experiences (ACEs) and DNA Methylation of Newborns in Cord Blood". In: *Clinical Epigenetics* 15.1 (2023), p. 162.

[40] Y. Fong, Y. Huang, D. Benkeser, L. N. Carpp, G. Áñez, W. Woo, A. McGarry, L. M. Dunkle, et al. "Immune Correlates Analysis of the PREVENT-19 COVID-19 Vaccine Efficacy Clinical Trial". In: *Nature Communications* 14.1 (2023), p. 331.

[41] Y. Fong, Y. Huang, B. Borate, L. W. P. van der Laan, W. Zhang, L. N. Carpp, I. Cho, G. Glenn, L. Fries, R. Gottardo, et al. "Antibody Correlates of Protection from Severe Respiratory Syncytial Virus Disease in a Vaccine Efficacy Trial". In: *Open Forum Infectious Diseases* 10.1 (2023), ofac693.

[42] C. B. Turley, L. K. Tables, T. Fuller, L. J. Sanders, H. Scott, A. Moodley, A. W. Davis, B. Leav, et al. "Modifiers of COVID-19 Vaccine Efficacy: Results from Four COVID-19 Prevention Network Efficacy Trials". In: *Vaccine* 41.33 (2023), pp. 4899–4906.

[43] **L. van der Laan**, W. Zhang, and P. B. Gilbert. "Nonparametric estimation of the causal effect of a stochastic threshold-based intervention". In: *Biometrics* 79.2 (2023), pp. 1014–1028.

[44] S. Daredia, K. Huen, **L. van der Laan**, P. A. Collender, J. C. Nwanaji-Enwerem, K. Harley, J. Deardorff, B. Eskenazi, N. Holland, and A. Cardenas. "Prenatal and Birth Associations of Epigenetic Gestational Age Acceleration in the Center for the Health Assessment of Mothers and Children of Salinas (CHAMACOS) Cohort". In: *Epigenetics* 17.13 (2022), pp. 2006–2021.

[45] Y. Fong, A. B. McDermott, D. Benkeser, S. Roels, D. J. Stieh, A. Vandebosch, M. Le Gars, G. A. Van Roey, et al. "Immune Correlates Analysis of the ENSEMBLE Single Ad26.COV2.S Dose Vaccine Efficacy Clinical Trial". In: *Nature Microbiology* 7.12 (2022), pp. 1996–2010.

[46] P. B. Gilbert, D. C. Montefiori, A. B. McDermott, Y. Fong, D. Benkeser, W. Deng, H. Zhou, C. R. Houchens, et al. "Immune Correlates Analysis of the mRNA-1273 COVID-19 Vaccine Efficacy Clinical Trial". In: *Science* 375.6576 (2022), pp. 43–50.

[47] **L. van der Laan**, A. Cardenas, R. Vermeulen, R. P. Fadadu, A. E. Hubbard, R. V. Phillips, L. Zhang, C. Breeze, W. Hu, C. Wen, et al. "Epigenetic Aging Biomarkers and Occupational Exposure to Benzene, Trichloroethylene and Formaldehyde". In: *Environment International* 158 (2022), p. 106871.

[48] A. Cardenas, R. P. Fadadu, **L. van der Laan**, C. Ward-Caviness, L. Granger, D. Diaz-Sanchez, R. B. Devlin, and M.-A. Bind. "Controlled Human Exposures to Diesel Exhaust: A Human Epigenome-Wide Experiment of Target Bronchial Epithelial Cells". In: *Environmental Epigenetics* 7.1 (2021), dvab003.

[49] J. C. Nwanaji-Enwerem, F. F.-L. Chung, **L. van der Laan**, A. Novoloaca, C. Cuenin, H. Johansson, B. Bonanni, A. E. Hubbard, M. T. Smith, S. J. Hartman, et al. "An Epigenetic Aging Analysis of Randomized Metformin and Weight Loss Interventions in Overweight Postmenopausal Breast Cancer Survivors". In: *Clinical Epigenetics* 13.1 (2021), p. 224.

[50] J. C. Nwanaji-Enwerem, **L. van der Laan**, E. F. Avakame, K. A. Scott, H. H. Burris, and A. Cardenas. "Associations of DNA Methylation Mortality Risk Markers with Congenital Microcephaly from Zika Virus: A Study of Brazilian Children Less than 4 Years of Age". In: *Journal of Tropical Pediatrics* 67.1 (2021), fmab020.

[51] J. C. Nwanaji-Enwerem, **L. van der Laan**, K. Kogut, B. Eskenazi, N. Holland, J. Deardorff, and A. Cardenas. "Maternal Adverse Childhood Experiences before Pregnancy Are Associated with Epigenetic Aging Changes in Their Children". In: *Aging* 13.24 (2021), p. 25653.

[52] J. C. Nwanaji-Enwerem, U. Nwanaji-Enwerem, **L. van der Laan**, J. M. Galazka, N. S. Redeker, and A. Cardenas. "A Longitudinal Epigenetic Aging and Leukocyte Analysis of Simulated Space Travel: The Mars-500 Mission". In: *Cell Reports* 33.10 (2020).

Reference Works

[53] D. Frauen, V. Melnychuk, **L. van der Laan**, and S. Feuerriegel. "Machine Learning for Causal Inference". In: *Wiley StatsRef: Statistics Reference Online*. Ed. by N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri, and J. L. Teugels. John Wiley & Sons, 2026, pp. 1–17.

Software, Protocols, and Research Materials

[54] P. B. Gilbert, Y. Fong, D. Benkeser, N. S. Hejazi, B. R. Borate, C. Yu, Y. Lu, **L. van der Laan**, et al. *COVID-19 Prevention Network Immune Correlates Analyses*. 2022.

[55] P. B. Gilbert, Y. Fong, D. Benkeser, J. Andriesen, B. Borate, M. Carone, **L. van der Laan**, et al. *CoVPN COVID-19 Vaccine Efficacy Trial Immune Correlates Statistical Analysis Plan*. 2021.

[56] J. R. Coyle, N. S. Hejazi, R. V. Phillips, L. W. P. van der Laan, and M. J. van der Laan. *hal9001: The scalable highly adaptive lasso*. 2020.

Talks and Presentations

Invited Oral Presentations

Online Causal Inference Seminar (OCIS), Young Researchers' Seminar, Online, 2026 – *Automatic Debiased Machine Learning for Functionals of M-estimands*

American Causal Inference Conference (ACIC), Invited Session: An Overview of Automatic Nonparametric (Riesz) Learning for Causal Effects Estimation, Salt Lake City, UT, 2026 – *Generalized autoDML for nonparametric M-estimands*

National Association for Business Economics (NABE), Seattle, WA, 2025 – *Novel Reinforcement Learning Models and Applications: Semiparametric Double Reinforcement Learning with Applications to Long-Term Causal Inference*
International Symposium on Biopharmaceutical Statistics (ISBS), Invited Session: Recent Advances in Highly Adaptive Lasso and Its Crucial Role in Causal Inference, Baltimore, MD, 2024 – *Adaptive Targeted Machine Learning of the Average Treatment Effect Using the Highly Adaptive Lasso*
Forum on the Integration of Observational and Randomized Data (FIORD), Washington, DC, 2023 – *Adaptive Debiased Machine Learning Using Data-Driven Model Selection Techniques*
International Conference on Computational and Methodological Statistics, Berlin, Germany, 2023 – *Adaptive Debiased Machine Learning with Model Selection*
Western North American Region of the International Biometric Society, Anchorage, AK, 2023 – *Nonparametric Inference on the Causal Effect of a Stochastic Threshold-Based Intervention*

Invited Short Courses

American Causal Inference Conference (ACIC), Seattle, WA, 2024 – *Highly Adaptive Lasso and Adaptive TMLE in Causal Inference* (*co-instructor with Mark van der Laan and Rachael Phillips*)

Contributed Oral Presentations

Joint Statistical Meetings (JSM), Topic-Contributed Paper Session, Boston, MA, 2026 – *Automatic Debiased Machine Learning for Smooth Functionals of Nonparametric M-Estimands*
American Causal Inference Conference, Detroit, MI, 2025 – *Automatic Double Reinforcement Learning in Semiparametric MDPs*
Conference on Causal Learning and Reasoning, Lausanne, Switzerland, 2025 – *Stabilized IPW via Isotonic Calibration* (*Best Paper*)
MIT Conference on Digital Experimentation, Cambridge, MA, 2024 – *Long-Term Causal Inference from Short-Term Experiments Using Reinforcement Learning*

Seminars

University College London Semiparametric Statistics Reading Group, Online, 2026 – *Adaptive Debiased Machine Learning Using Data-Driven Model Selection Techniques*
Université de Montpellier, Julie Josse Group, 2025
International Seminar on Selective Inference, 2024
Stanford Causal Inference Working Group, Stanford, CA, 2023
University of Pennsylvania Causal Inference Seminar, Philadelphia, PA, 2023

Contributed Posters

NeurIPS 2026, 2026 – *Calibeating Prediction-Powered Inference*
NeurIPS 2026, 2026 – *Inverse Reinforcement Learning with Just Classification and a Few Regressions*
Stanford Causal Science Center Student Conference, Stanford, CA, 2026 – *Grounded Causal Inference*
ICML 2025, Vancouver, BC – *Generalized Venn and Venn-Abers Calibration*
ACIC 2025, Detroit, MI – *Automatic Debiased Machine Learning for Smooth Functionals of Nonparametric M-Estimands*
ACIC 2025, Detroit, MI – *Automatic Doubly Robust Inference via Calibration*
NeurIPS 2024, Vancouver, BC – *Self-Calibrating Conformal Prediction*
ACIC 2024, Seattle, WA – *Combining T-learning and DR-learning: A Framework for Oracle-Efficient Estimation of Causal Contrasts*
ICML 2023, Honolulu, HI – *Causal Isotonic Calibration for Heterogeneous Treatment Effects*

Professional Service

Reviewer: JRSS-B, JASA, Annals of Statistics, Biometrika, Biostatistics, Journal of Applied Econometrics, EJS, JMLR, JCI, NeurIPS, ICLR, ICML

Session Organizer:
American Causal Inference Conference, 2024 – *Highly Adaptive Lasso and Adaptive TMLE in Causal Inference*
International Symposium on Biopharmaceutical Statistics, 2024 – *Recent Advances in Highly Adaptive Lasso and Its Crucial Role in Causal Inference*

Session Chair: American Causal Inference Conference, 2026

Invited Discussant: Online Causal Inference Seminar, 2026

Teaching Experience

University of Washington
Ph.D. Theory Sequence in Statistics (STAT 581, 582, 583), Teaching Assistant, 09/2022–06/2023
Wrote exams, led weekly sections, held office hours, graded assignments, and mentored graduate students.

University of California, Berkeley
Introduction to Probability and Statistics (STAT 201A, 201B), Personal Tutor for Biostatistics Group, 08/2019–05/2020

University of Groningen
Introduction to Mathematics for Biomedical Sciences, Teaching Assistant, 09/2018